

Artemis Search for Supernova Isotopes in the Lunar Regolith

Most supernova explosions in our Galaxy occur at great distances but, over long timescales, nearby explosions are likely. Events within ~ 10 pc (30 light-years) would endanger life on Earth (Ruderman 1974; Ellis and Schramm 1995) and should occur every few hundred million years. It was suggested in Ellis et al. (1996) that discovery of short-lived radioactive isotopes such as ^{60}Fe would be direct evidence for near-Earth supernova explosions, which might have affected the Earth's biosphere.

Following this suggestion, live, i.e., undecayed, radioactive ^{60}Fe (half-life $t_{1/2} = 2.6$ Myr) has been detected by multiple experimental teams in deep-ocean samples around the world (Knie et al. 1999, 2004; Fitoussi et al. 2008; Wallner et al. 2016; Ludwig et al. 2016), in cosmic rays (Binns et al. 2016), and in Antarctic snow (Koll et al. 2019). Importantly, ^{60}Fe has also been found in *Apollo* drive tube samples of the lunar regolith (Fimiani et al. 2016), confirming that the radioisotope deposition was not limited to Earth but prevalent in the solar system. The *Apollo* study showed that the $^{60}\text{Fe}/^{53}\text{Mn}$ ratio exceeds that due to lunar bombardment by cosmic rays (and that found in meteorites), confirming that the ^{60}Fe exceeds the expected cosmogenic production. These discoveries provide compelling evidence that at least one supernova exploded within ~ 100 parsecs of the Earth (Fields & Ellis 1999; Fields et al. 2008; Fry et al. 2016) within the past 3 million years.

Artemis has a unique scientific opportunity to confirm the supernova interpretation of the ^{60}Fe data, to clarify the behavior of dust in supernova remnants and possibly the direction of the recent parent supernova and, by searching for other live isotopes such as ^{53}Mn , ^{146}Sm and ^{244}Pu , could identify what type of supernova or kilonova exploded within the vicinity of Earth.

Specifically, we call for *Artemis* to perform the following experiments on lunar regolith:

- Return well-stratified regolith samples from depths down to at least ~ 1 meter.
- Return regolith samples expected to exhibit different rates of gardening.
- Collect samples from under boulders, to serve as controls lacking the recent supernova signal.
- Take close-up photographs of the undisturbed regolith with the highest possible resolution.

Our science can piggyback on other lunar regolith sample return studies. Upon return, the trace radioisotope compositions can be characterized via Accelerator Mass Spectrometry (AMS) measurements on small subsamples. Fimiani et al. (2016) used AMS to discover ^{60}Fe in lunar soil samples, with 107 counts found. Recent improvements in efficiency will yield a few 100 ^{60}Fe detector events per gram of lunar soil, which is 10 to 100 times above background. Thus only a few grams are needed at each site. The radioisotope profile (Fimiani et al. 2016) can thus be measured precisely as a function of depth, and it is expected that mixing and turnover are important to depths ~ 1 meter (Costello et al. 2018). Measurements for different gardening rates should lead to different mixing and thus different radioisotope abundances, thereby probing the time and type of supernova debris injection. Conversely, the time-resolved terrestrial ^{60}Fe data could be used as calibration, enabling the ^{60}Fe profile on the Moon to serve as a novel probe of gardening effects and their time history.

Supernova-manufactured ^{60}Fe nuclei reach the Earth and Moon in dust particles Fry et al. (2020). They may arrive directly or after deflection by magnetic fields. In the former case, comparing *Artemis* measurements near the lunar southern pole with *Apollo* measurements near the equator (and/or other future measurements) could probe the direction of the parent supernova, as illustrated in the Figure.

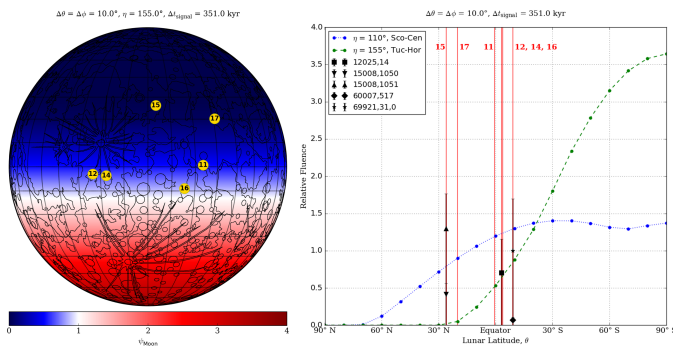


Figure 1. Left: latitude dependence of ^{60}Fe in the lunar regolith, assuming direct arrival from a supernova in the Tucana-Horologium star cluster; yellow circles are Apollo landings. Right: ^{60}Fe compared for Tucana-Horologium and Scorpio-Centaurus star clusters; the vertical bars are Apollo data.

Detection of either of these isotopes would be evidence for one or more explosions within the past few hundred million years. Supernovae may not produce ^{244}Pu , but it is expected to be produced by kilonovae, so a signal in the lunar regolith, where there is no anthropogenic background, would be particularly interesting. Here we expect ~ 5 detector events per sample gram, but with no lunar background, so that any detection points to an extraterrestrial source.

Searches for the above-mentioned isotopes would provide, with minimal investment, a high-impact Artemis science program that would provide information about explosive nearby events in the Galaxy, shedding light on the origins of heavy elements and possible long-term risks to life on Earth.

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The two panels show the latitude dependence expected for ^{60}Fe propagating directly from a supernova for two proposed origin sites; the ^{60}Fe latitude profile can distinguish these.

Other radioisotopes are of interest. ^{53}Mn ($t_{1/2} = 3.7$ Myr) was not found in *Apollo* cores (Fimiani et al. 2016), but it has been reported in deep-ocean samples (Korschinek et al. 2020). *Artemis* data may be able confirm this discovery; measurements of the relative amounts of ^{60}Fe and ^{53}Mn could distinguish between different supernova progenitors.

Two interesting longer-lived isotopes are ^{146}Sm ($t_{1/2} = 68$ Myr) and ^{244}Pu ($t_{1/2} = 81$ Myr), and ^{244}Pu has been reported in deep-ocean sediments (Wallner et al. 2015).